LIF equivalent circuit (portfolio)

Leaky integrate-and-fire (LIF) equivalent circuit - matches alexwoods1.github.io
Source: data/lif-circuit.json - regenerate with scripts/generate_circuit_pdf.py

Membrane

C = 200 pF
R = 100 MOhm
tau_m = 20 ms
tau_m dV/dt = -(V - V_rest) + R·I

Voltages

V_rest: -65 mV
V_th: -50 mV
V_reset: -70 mV
V_spike_peak: 30 mV

Comparator

Texas Instruments TLV1701DBVR (SOT-23-5)
Role: Threshold detector (V >= V_th triggers spike output)
supply_V: 2.2 to 36 (use +/-9 V dual supply for headroom)
input_offset_max_mV: 0.55
propagation_delay_ns_typ: 260
output: Push-pull, rail-to-rail near supply
threshold_behavior: Non-inverting input = V_membrane; inverting input = V_th reference

Parts list

R1 100 MOhm Leak resistor (membrane)
  Example: Ohmite HVC0805T1005MJE or 100 MOhm 1% film (series if needed)

C1 200 pF Membrane capacitance
  Example: Murata GRM1885C1H221JA01D (220 pF NP0) or 200 pF C0G 1%

U1 TLV1701 Comparator / threshold
  Example: TI TLV1701DBVR

U2 OP07 Precision buffer for V_rest reference
  Example: TI OP07CDR

U3 OPA2333 Howland-style current source for I(t)
  Example: TI OPA2333AIDGKR

SW1 SPST Reset switch (pull V to V_reset)
  Example: C&K OS102011MA1Q1 or tactile NO

V_rest -65 mV Rest reference
  Example: Precision divider from -9 V using OP07 buffer (scale per bench calibration)

V_th -50 mV Threshold reference
  Example: Trimmer + precision divider into TLV1701 inverting input

V_reset -70 mV Reset level
  Example: SPST network to reset node when spike ends

Spike output

Piecewise linear pulse (site uses symbolic +30 mV peak)
Physical breadboard version: TLV1701 output drives astable or logic-level pulse; portfolio spike path is decorative